

Development of *Beauveria bassiana* dry formulation for vectoring by honey bees *Apis mellifera* (Hymenoptera: Apidae) to the flowers of crops for pest control

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Abstract

Using bee pollinators as a means for the dissemination of microbial control agents, such as *Beauveria bassiana*, against insect pests of agricultural crops is a novel and interesting approach to biological control. In four laboratory trials, one in Canada and three in Jordan, factors affecting the acquisition of *B. bassiana* by honey bees were evaluated using hive-mounted inoculum dispensers. The numbers of conidia carried by bees emerging from the dispensers differed according to the type of carrier used. Bees that passed through corn flour acquired more inoculum than did those that walked through wheat flour, durum semolina, corn meal, potato starch, potato flakes, oat flour or barley flour. The numbers of conidia acquired by the bees increased with decreasing particle size and moisture content of the carrier, and with increasing density of *B. bassiana* conidia in the formulation. Time required for a bee to pass through the dispenser did not significantly affect the acquisition of conidia. This study indicated that honeybees (*Apis mellifera carnica*) have a great potential for vectoring *B. bassiana* in crop systems. It also opens more avenues for studies on bee delivery of other microbial biological control agents.

Keywords: *Apis mellifera*, *Beauveria*, bee vectoring, dispensers, formulation

Introduction

Fungal species such as the mitosporic fungi, *Beauveria bassiana* (Balsamo) Vuillemin and *Metarhizium anisopliae* (Metschnikoff) exhibit wide host ranges that include many insect pest species. Those agents can be produced on inexpensive artificial media and have long shelf lives. *Beauveria bassiana* has been used commercially for pest control worldwide for more than 25 years (Goettel et al. 1990) and in 1998 was first registered in the USA for the control of a wide range of insects on different crops.

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Successful use of fungi as microbial control agents depends ultimately on the use of the right propagule, formulated in an appropriate manner, and applied at an appropriate dosage and time (Lacey et al. 2001). Infective propagules of entomopathogenic fungi display great variation in size, dryness, and surface hydrophobicity (Boucias & Pendland 1991). Fungal conidia with hydrophobic cell walls, such as *B. bassiana*, *M. anisopliae*, and *Nomuraea rileyi* (Farlow) Samson, are difficult to suspend uniformly in water. Moreover, because many entomopathogenic fungi infect their hosts through the cuticle, optimal coverage during application in the field is a key factor for success. For example, poor control of *Lygus* bugs by *B. bassiana* in alfalfa seed fields was attributed to insufficient spray penetration within the plants' canopy (Noma & Strickler 2000). In addition, some application methods, such as mechanical sprays, adversely affect viability and virulence of the entomopathogenic fungi used. The viability of both blastospores and conidia of *Verticillium lecanii* (Zimmermann) Viégas decreased with the duration of the pumping period and pressure in a high-pressure hydraulic sprayer (Nilsson & Gripwall 1999). In addition, the viability of oil-formulated conidia of *M. anisopliae acridum* (as flavoviride) was reduced by 30% after passage through an exhaust nozzle sprayer; the adverse effect attributed to brief exposure of the conidia to temperatures over 100°C (Griffiths & Bateman 1997).

Novel technology for the application of microbial control agents is the use of pollinating bees (honeybees (*Apis mellifera* L.) and bumble bees (*Bombus* spp.) (Hymenoptera: Apidea)). The first trials using this technology were made in the early 1990s with the dissemination of fungi and bacteria against plant pathogens and pests (review in Kevan et al. 2003, 2006). In all these experiments, dry formulations of the antagonists were placed in special hive-mounted dispensers that allowed the bees to acquire the biological control agent as they passed through the inoculum. Formulating the biological control agents' infective propagules as a mixture has many advantages: the propagules are not easily blown out of the dispensers by air currents produced by the wing strokes of the bees; the technique also extends the use of the propagules protected in the hive-mounted dispensers, reduces the need for application of large amounts of the agent so making the technology cost effective, and results in highly site-specific delivery to the flower and plant surface. Despite successes reported previously, practicality necessitates refinement.

In previous reports of using bees as vectors of microbial biological control agents (Kevan et al. 2003, 2006), two approaches were taken for preparation of inoculum mixtures for use in hive-mounted dispensers. First, the propagules of the biological control agent are applied to a suitable substrate which, when colonized, is dried, pulverized and screened to eliminate large particles for use. The resulting powder is then formulated with a carrier/diluent (Peng et al. 1992; Yu & Sutton 1997). Second, commercially available inoculum is prepared for use by mixing it with a carrier/diluent (e.g. Butt et al. 1998; Kovach et al. 2000). Both approaches allow for the preparation of different concentrations of the inocula, which can be important for the management of different pests, but the latter is less labor intensive. The nature of the carrier/diluent is important (Kevan et al. 2006). Israel and Boland (1993) noted that some carriers, notably talc, seem to irritate honey bees which then groom themselves to remove varying amounts of the formulation. Butt et al. (1998) used costly laboratory grade Styrofoam particles (Biobeads, particle size 40–80 µm; Bio-Rad, Hercules, CA, USA). All carriers that we used are inexpensive, easily available, and among the least irritating to honey bees (Israel & Boland 1993).

Further, when choosing a diluent to mix with the propagules of an entomopathogen for bee vectoring, it is important to consider the ability of bees to acquire the inoculum, the risk to the bee vectors' health (Kevan et al. 2006) and the cost of the commercially available form of the agent. Our study evaluates different materials as possible diluents for *B. bassiana* spores for honey bee delivery. Factors influencing bee loads (doses) of the fungal spores including diluent particle size, concentration of infective propagules in the inoculum, moisture content of the diluent and time spent by bees to crawl through an inoculum dispenser were part of our study.

Materials and methods

Experimental material

Non-formulated dry conidia of *B. bassiana* Strain GHA were provided by Emerald BioAgriculture Corp, Salt Lake City, UT. Conidia were kept at 4°C until used for preparation of the inocula. Shortly before the experiments, and to assess the number of colony forming units (cfu) per unit weight of conidia, six 0.1-g samples were taken at random, suspended in 100 mL sterile distilled water plus 0.1% Tween 20 and agitated at 110 rpm for 2 h on a rotary shaker. Three aliquots of 0.1 mL of 10-fold serial dilutions of the aqueous suspensions were placed onto oatmeal agar medium (Difco, Detroit, MI) amended with 0.55% Dodine, 0.005% chlortetracycline and 0.01% crystal violet in Petri dishes (Chase et al. 1986). The dishes were kept in darkness at $22 \pm 1^\circ\text{C}$ for 4–5 days, after which colonies of *B. bassiana* were counted and recorded.

Hive mountable inoculum dispensers, similar to those used by Peng et al. (1992), were constructed and adapted for our laboratory trials. The dispenser consisted of a rectangular wooden box ($25 \times 12 \times 4$ cm) with a honey bee entrance at one end and into which a Perspex tray ($20 \times 8 \times 1$ cm) could be inserted after being filled with the formulation. The dispenser had an exit hole at the opposite end from which honey bees could be collected after passing through the formulation. Honey bees (*Apis mellifera carnica* (Buckfast strain)) were used in the trials in Canada, and *A. mellifera* (hybrid between *syriaca* and *italica*) was used for the trials in Jordan. The Canadian bees were collected from honey bee hives at the University of Guelph apiary. Those used in Jordan were collected from a private apiary. They were brushed from the frames into a wooden cage ($10 \times 15 \times 8$ cm height) and used the same day. We used 25 g of formulation in each experiment. Each of the following experiments was done 1 time using 10 bees passing through one dispenser in Canada, and again, in exactly the same way three times using 30 bees, passing through three different, but identical, dispensers in Jordan.

Carrier types

The following commonly and commercially available materials were evaluated as carriers: graham cracker crumbs, wheat flour, durum semolina, corn meal, corn flour, potato starch, potato flakes, oat flour and barley flour. Carriers were prepared by mixing 100 mL distilled water with 100 g of each carrier. The mixtures were autoclaved for 20 min at $121 \pm 1^\circ\text{C}$ and 1 atmosphere pressure. The mixtures were then placed in an oven at 125°C for 7 days until completely dry as determined by gravimetry. Carriers were then cooled and mixed with *B. bassiana* conidia to a

concentration of 5×10^8 colony forming units (cfu)/g. Twenty-five grams of each carrier were placed in the dispenser's Perspex tray, leveled, and the tray inserted into the dispenser. Ten honey bees were allowed to pass through each carrier. Each bee entered at one end of the dispenser (at time 0) and was collected and timed as it emerged from the exit. To determine the density of *B. bassiana* conidia on individual bees, each collected bee was individually placed into a 250-mL Erlenmeyer flask with 100 mL distilled water plus 0.1% Tween 20 and agitated at 110 rpm for 2 h on a rotary shaker. Densities of *B. bassiana* conidia in the aqueous suspensions in each flask were determined by serial dilution plating as described above.

Carrier particle size

The effect of particle size of the carrier on acquisition of *B. bassiana* by honey bees was evaluated using corn flour, durum semolina and glass beads. These carriers were selected based on results of the 'carrier type' experiment (above). The carriers were sifted to particles of three different sizes: 45–90, 90–150 and 150–300 μm and were prepared as above. Each carrier particle size was mixed with *B. bassiana* conidia to a concentration of 5×10^8 cfu/g and was tested for bee acquisition as above. The time required by each bee to pass through the dispenser with each carrier particle size was recorded.

Inoculum concentration

Based on the experimental results with carrier type (above) and particle size (above), corn flour of particle sizes ranging from 45 to 90 μm was selected to study the relationship between concentration of conidia in the formulation and acquisition of the conidia by honey bees. The carrier was prepared as described above and mixed with *B. bassiana* conidia to give four concentrations: 1×10^6 , 1×10^7 , 1×10^8 and 1×10^9 cfu/g. The four concentrations were tested in the dispensers as described above. The time required by each bee to pass through each concentration was recorded.

Inoculum moisture content

Based on the experimental results of carrier particle sizes (above), corn flour of particle sizes ranging from 45 to 90 μm was also selected to study the impact of moisture content of the formulation on bee acquisition. Corn flour was prepared and autoclaved as above. After it was dried completely by placing it in an oven at 125°C for 7 days (above), three treatments were prepared: dry corn flour, semi-dry corn flour which was prepared by adding 25% wt/wt sterile distilled water to dry corn flour, and wet corn flour which was prepared by adding 50% (w/w) sterile distilled water to dry corn flour. Each treatment was mixed with *B. bassiana* conidia to a concentration of 5×10^8 cfu/g and then was tested for bee acquisition as above. The time required by each individual bee to pass through each treatment was recorded.

Statistical analysis

PROC univariate, residual analysis was applied to test if the data met the assumptions of analysis of variance (ANOVA). When assumptions were not met, the data were

transformed using natural logarithm. The data obtained from the trials in Canada and Jordan were compared using ANOVA to detect the possibility of combining trials. In all the trials, the results from each 10 bees passing through one dispenser were averaged and statistical analysis was done on the averages. The experiment on carrier type was analyzed as a one way ANOVA and the means compared by using Student–Newman–Keuls (SNK) multiple-range test. Experiments on carrier particle size and moisture content were analyzed as covariance for two ways ANOVA and one way ANOVA, respectively, with the time as the co-variable. If interactions between variables were not significant, the main effects of the variables were examined by comparing means using SNK multiple-range test. The experiment on inoculum concentration was analyzed by stepwise regression analysis. Type I error (α) was 0.05 for all tests and means for all tests were compared only when the *F*-test was significant ($P < 0.05$). All statistical analyses were made by statistical analysis system software (SAS) version 8, (SAS Institute 1999).

Results

Analysis of variance showed no significant differences between trials done in Canada and Jordan for carrier type ($F_{3,21} = 2.7$, $P > 0.07$), carrier particle size ($F_{3,15} = 0.37$, $P > 0.77$), inoculum concentration ($F_{3,12} = 0.1$, $P > 0.96$) and inoculum moisture content ($F_{3,3} = 8.3$, $P > 0.06$) so the results from all four trials were combined.

Carrier type

All bees that passed through the carriers acquired large amounts of *B. bassiana* conidia. Statistical analysis showed significant differences among the different substrates tested in the experiment ($F_{7,21} = 8.5$, $P > 0.01$). Further mean separation showed that bees that passed through dispensers with corn flour carried significantly more conidia than did bees that passed through dispensers with graham crumbs, corn meal, and potato flakes, but did not differ significantly, from those that passed through dispensers with durum semolina, oat flour, potato starch, and wheat flower (Table I).

Table I. Mean ($\pm$ SE) density of colony forming units (cfu) of *Beauveria bassiana* per bee detected on honey bees that crawled through various carriers at 5×10^8 cfu/g of carrier and ranked according to density from highest to lowest.

Carrier type	Mean ($\pm$ SE) Log _e cfu/bee cfu/bee
Corn flour	14.2 (0.33) a* 1.7×10^6
Durum semolina	13.7 (0.47) ab 1.2×10^6
Wheat flour	13.5 (0.18) ab 8.8×10^5
Oat flour	13.4 (0.32) ab 7.5×10^5
Potato starch	13.3 (0.23) ab 6.5×10^5
Corn meal	12.5 (0.32) bc 3.1×10^5
Graham cracker crumbs	12.2 (0.22) c 2.0×10^5
Potato flakes	12.0 (0.24) c 1.8×10^5

*Means within columns with different letters differ significantly at 0.05 level using Student–Newman–Keuls (SNK) multiple-range test. Number of trials $N = 4$. Total number of honey bees used for each carrier $n = 40$.

Carrier particle size

Analysis of co-variance indicated that the effect of time spent by bees crawling through carriers on bee loads of conidia of *B. bassiana* was not significant at 0.05 level ($F_{1,15} = 0.01$, $P > 0.92$). Moreover, the interactions between time and carrier type ($F_{2,15} = 0.55$, $P > 0.59$), time and carrier particle size ($F_{2,15} = 1.02$, $P > 0.38$), and carrier particle size and carrier type ($F_{4,15} = 1.04$, $P > 0.42$) were not significant at 0.05 level. On the other hand, significant differences occurred among the substrates ($F_{2,15} = 47.04$, $P > 0.01$) and among the particle sizes ($F_{2,15} = 10.26$, $P > 0.01$).

Means of cfu on honey bees that crawled through carriers of different particle sizes were pooled over the carrier types because the interaction effect between carrier type and carrier particle size was not significant. Mean cfu/bee of carriers with particle size 45–90 μm was significantly higher than mean particle sizes 90–150 and 150–300 μm . In addition, mean cfu/bee of particle size 90–150 μm was significantly higher than mean particle size 150–300 μm (Table II). Means of cfu on honey bees that crawled through carriers of different particle sizes pooled over particle size are not presented because the results were similar to Table I above.

Inoculum concentration

Stepwise regression analysis indicated that the variable, inoculum concentration, should be included in the model ($F_{1,14} = 115.59$, $P < 0.01$) but that time spent by bees passing through the inoculum should not. Thus, the best-fit regression equation (Figure 1) to predict the amount of *B. bassiana* conidia on bees as they crawled through the dispenser was:

$$\text{Log cfu/bee} = -0.50 + 0.75(\text{log concentration of conidia}) (R^2 = 0.89).$$

Inoculum moisture content

Density of cfu on honey bees that crawled through corn flour 45–90 μm varied with the different moisture contents (Table III). Bees that crawled through dry corn flour carried significantly more *B. bassiana* conidia than did bees that crawled through wet corn flour. No significant differences were found between bees that crawled through dry corn flour and semi-dry corn flour (Table III).

Table II. Mean ($\pm$ SE) colony forming units (cfu) of *Beauveria bassiana* per bee detected on honey bees that crawled through three carrier particle sizes pooled over carrier type at 5×10^8 cfu/g of carrier.

Particle size	Mean ($\pm$ SE)	
	Log _e cfu/bee cfu/bee	
45–90 μm	13.5 (0.23) a*	8.3×10^5
90–150 μm	12.9 (0.26) b	5.1×10^5
150–300 μm	12.4 (0.33) c	3.3×10^5

*Means within columns with different letter differ significantly at 0.05 level using Student–Newman–Keuls (SNK) multiple-range test. Number of trials $N = 4$. Total number of honey bees used for each size $n = 40$.

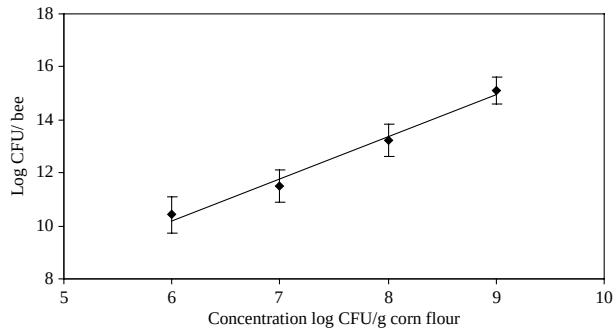


Figure 1. Effect of inoculum concentration of *B. bassiana* on honey bee acquisition of conidia of *B. bassiana* after bees were passed through fine corn flour carrier with four concentrations ranging between 1×10^6 and 1×10^9 colony forming units (cfu) per g carrier. Axes are by natural logarithms.

Discussion

Honey bees that walked through dispensers filled with inocula of *B. bassiana* formulated with different carriers acquired large doses of conidia. Those conidia, when deposited on the flowers or leaves of crops, have potential for controlling pests that feed on, or inhabit flowers, such as thrips (Thysanoptera) and *Lygus* spp. (Hemiptera) (Al-Mazra'awi et al. 2006a). Nevertheless, commercially available inocula should be diluted before application to assure cost-effectiveness, efficacy, and safety to insect vectors, such as honey bees (Kevan et al. 2003, 2006).

The density of conidia of *B. bassiana* acquired by the bees differed significantly in relation to the carriers (that also are diluents) used in the inoculum formulation (Table I). In general, bees that crawled through flours and starch carried more conidia than did those that crawled through crumbs and flakes. These differences can be attributed to the differences in particle sizes of the different carriers, their specific gravity and particle configuration. Our experiment on particle size confirmed that when honey bees crawled through inocula formulated with diluent/carriers of particle sizes 45–90 μm , they acquired significantly more conidia than those that crawled through formulations with larger particle sizes of the three tested carriers (Table II). The smaller particles have more surface area per unit mass and, as a result, may provide relatively more adhesion sites for the conidia. Moreover, smaller particles have less mass than bigger ones and this might also facilitate their acquisition and transport by

Table III. Mean ($\pm$ SE) colony forming units (cfu) of *Beauveria bassiana* per bee detected on honey bees that crawled through carrier formulations of dry corn flour (no evaporable moisture), semi-dry corn flour (25%, w/w, sterile distilled water added to dry corn flour), and wet corn flour (50%, w/w, sterile distilled water added to dry corn flour) at 5×10^8 cfu/g of carrier.

Carrier	Mean ($\pm$ SE) Log _e cfu/bee
Dry corn flour	14.2 (0.26) a* 1.9×10^6
Semi-dry corn flour	13.9 (0.27) a 1.3×10^6
Wet corn flour	12.4 (0.27) b 3.3×10^5

*Means within columns with different letter differ significantly at 0.05 level using Student–Newman–Keuls (SNK) multiple-range test. Number of trials $N=4$. Total number of honey bees used for each moisture content $n=40$.

bees. Our finding also may explain the differences found in acquisition of conidia of *B. bassiana* when mixed with corn flour, durum semolina or glass beads. Glass beads are denser, heavier, and smoother than particles of corn flour or durum semolina making them harder to be picked up by the bees. Although particle configuration was not studied in these experiments, microscopic examination of corn flour and durum semolina showed that both particles have irregular shapes, but glass beads are more uniform and spherical.

Inoculum concentration in the tested formulations affected the amount of *B. bassiana* conidia acquired by honey bees. A positive linear relationship was established between the concentration of conidia in the dispenser and bee body loads. This finding may be important for applying different amounts of inocula according to the susceptibility of the target pest and safety to the vector.

Our results showed that inoculum moisture content affects the numbers of conidia acquired by honey bees. The drier the inoculum, the more conidia were picked up. Wet carriers form a crust that allows the bees to crawl over the carrier without becoming dusted with conidia. Moreover, wet particles are heavier than dry ones and that too may reduce their acquisition by crawling bees.

In experiments with particle sizes, conidial density and formulation moisture content time was studied as a covariant. In all experiments, time spent by the bees to crawl through the inoculum had no effect on bee body loads of conidia. The shortest time spent by a bee crawling through the dispensers in these experiments was 17 s. Bees become dusted with inoculum after crawling for only a few seconds (<5 s), after which time the body becomes saturated with the inoculum.

Improving bee delivery requires the refinement of suitable formulations to facilitate dispersal of microbial control agents to crops. We analyzed some factors that affect the acquisition of *B. bassiana* conidia by honey bees and found that these factors influenced the efficiency of bee delivery. We note that at the doses required for pest treatment, risks to bees are minimal (Al-Mazra'awi et al. 2006b, Kevan et al. 2006) but trials involving beneficial non-vectors, such as other pollinators, need to be made. In addition, refined delivery technology may open more avenues for research using honey bees, bumble bees and other insects as vectors of microbial control agents against agricultural pests.

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